433 Project - PART 1

Didem Ugur - 2375962
Baris Akdag - 2574457

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1 Introduction

Active Noise Cancellation (ANC) is a signal processing technique designed to reduce unwanted acoustic disturbances. It works by generating an “anti-noise” signal with the exact opposite phase at the cancellation point. Compared to passive isolation methods (like standard earcups), ANC is particularly effective at targeting low-frequency noise where mechanical attenuation is often limited.

In this project, offline implementations of ANC are studied across two distinct configurations:

- **Case 1 (Feedforward ANC):** A single external microphone captures the incoming disturbance *before* it reaches the ear. The system estimates the disturbance at the listening point and generates an inverse waveform to cancel it.
- **Case 2 (Feedback + Feedforward ANC):** A second “error” microphone is placed inside the headphone to measure the residual signal at the ear. This provides a direct error signal for adaptation, closely mirroring a practical, real-world ANC system.

Objective: The primary goal of Part 1 is to implement both cases offline in MATLAB and demonstrate successful noise suppression (aiming for > 10 dB) using objective performance measures. Because ANC performance is heavily dependent on the nature of the noise, the algorithms are evaluated on several disturbance types, spanning both narrowband and broadband examples.

This report utilizes the **NLMS** algorithm for Case 1 and the **FxNLMS** algorithm for Case 2 (to account for the secondary path between the controller output and the listening point). A comparison with an NLMS baseline is also included for Case 2. Performance is evaluated via suppression levels, signal-to-noise ratio (SNR) improvement, and output error.

2 Problem Definition

The objective is to suppress the disturbance reaching the listening point while preserving the desired audio signal (e.g., music). Since Part 1 is conducted offline, the acoustic propagation of the disturbance is simulated in MATLAB rather than measured in real-time.

- **Primary Path $P(z)$:** The disturbance at the ear is generated by passing recorded reference noise through a simulated FIR filter. This models the acoustic propagation (delays, simple reflections) from the external microphone to the ear.
- **Secondary Path $S(z)$ (Case 2 Only):** The anti-noise generated by the adaptive filter is also propagated through a second FIR filter. This models the physical path between the headphone speaker (controller output) and the ear/internal microphone.

2.1 Case 1: Feedforward ANC

A single external reference microphone captures the incoming noise. The adaptive filter processes this reference signal to generate an anti-noise cancellation signal. Because there is no internal microphone to provide hardware feedback, the offline study simulates the ear's disturbance by filtering the reference noise through the primary-path FIR model $P(z)$. Performance is evaluated by comparing the noisy and cleaned signals offline.

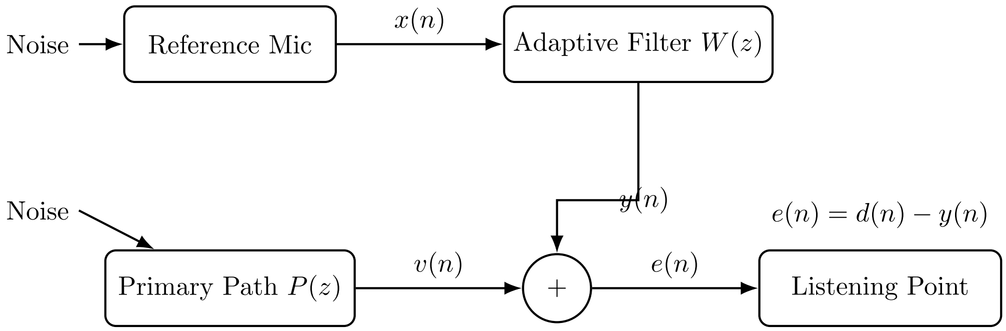


Figure 1: Block representation of the offline Case 1 feedforward ANC model.

2.2 Case 2: Feedback + Feedforward ANC

This setup utilizes two microphones: the external reference microphone and an internal error microphone inside the headphone. The internal microphone measures the residual signal at the ear, allowing the controller to refine the adaptive filter.

In this offline simulation, the primary-path FIR filter generates the disturbance, while the anti-noise passes through the secondary-path FIR filter. Consequently, the error signal is dependent not just on the adaptive filter's output, but also on how the secondary path alters that output. The computational workflow for this offline implementation is illustrated in Figure 2.

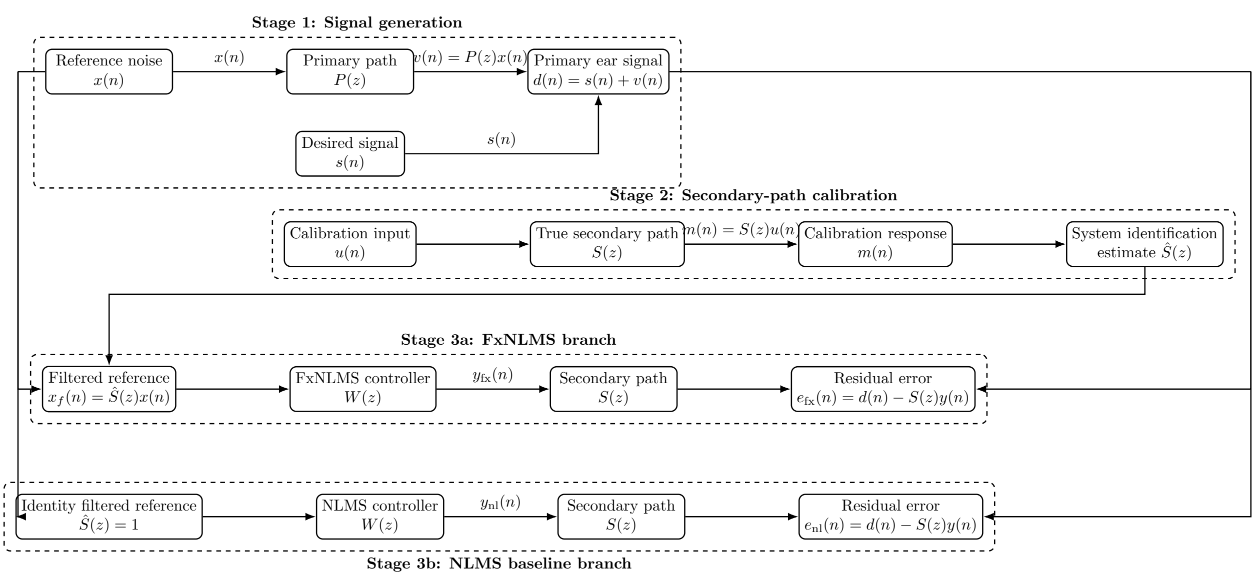


Figure 2: Workflow of the offline Case 2 implementation. The disturbance is first generated through the primary-path FIR model. The secondary path is identified through a calibration stage, and the estimated model is then used in the FxNLMS branch. For fair comparison, the NLMS baseline uses an identity $\hat{S}(z)$ in the update, while the same true secondary path $S(z)$ is retained in the ear-error computation.

2.3 Evaluation Criteria

The project requires demonstrating a disturbance suppression of **more than 10 dB**, supported by output SNR and output error metrics. Since structured/narrowband noises (like single tones) are easier to suppress than unstructured/wide-band disturbances (like white noise), the algorithms are tested across multiple signal characteristics to provide a comprehensive evaluation.

3 Theoretical Background

3.1 Adaptive Filtering Fundamentals

Adaptive filters are essential when the system or disturbance characteristics are unknown or change over time. In ANC, the goal is to generate an anti-noise signal that minimizes the **mean squared error (MSE)** between the desired signal and the filter output.

For an adaptive filter with coefficient vector $\mathbf{w}(n)$ and input vector $\mathbf{x}(n)$, the filter output is:

$$y(n) = \mathbf{w}^T(n)\mathbf{x}(n) \tag{1}$$

The residual error signal is:

$$e(n) = d(n) - y(n) \tag{2}$$

where $d(n)$ is the primary signal at the cancellation point. The cost function to minimize is $J(n) = E[e^2(n)]$.

Least Mean Squares (LMS):
The standard LMS algorithm updates the coefficients in the negative gradient direction of the instantaneous squared error:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu e(n)\mathbf{x}(n) \tag{3}$$

where μ is the step size. However, LMS performance fluctuates with input signal power.

Normalized LMS (NLMS):
To improve stability, NLMS scales the update by the input signal's energy:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\mu}{\delta + \mathbf{x}^T(n)\mathbf{x}(n)} e(n)\mathbf{x}(n) \tag{4}$$

where δ is a small positive constant for numerical stability. This makes it ideal for ANC, where reference noise levels vary.

3.2 Primary and Secondary Paths

- **Primary Path $P(z)$:** The transfer from the noise source to the ear. The disturbance is modeled as $v(n) = P(z)x(n)$. With a desired audio signal $s(n)$, the primary signal at the ear is $d(n) = s(n) + v(n)$.
- **Secondary Path $S(z)$:** The transfer from the adaptive filter output to the ear. The actual anti-noise arriving at the ear is $y_s(n) = S(z)y(n)$.

Therefore, the true residual error in a practical ANC system is:

$$e(n) = d(n) - S(z)y(n) \tag{5}$$

Filtered-x LMS (FxLMS):
In Case 2, ignoring $S(z)$ means the adaptation direction won't match physical reality. To fix this, the reference signal is first filtered through an estimate of the secondary path, $\hat{S}(z)$:

$$x_f(n) = \hat{S}(z)x(n) \tag{6}$$

The FxNLMS algorithm then uses this filtered reference for the weight update:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\mu}{\delta + \mathbf{x}_f^T(n)\mathbf{x}_f(n)} e(n)\mathbf{x}_f(n) \tag{7}$$

4 Experimental Setup

Both cases follow a shared MATLAB pipeline. It is important to reiterate that all acoustic propagations—including both the primary path for the disturbance and the secondary path for the anti-noise in Case 2—are simulated offline strictly using discrete FIR models. The general workflow is as follows:

1. Load the recorded reference noise and clean desired audio signal.
2. Propagate the noise through the simulated primary path $P(z)$ to generate ear-level disturbance.
3. Add the desired signal to create the primary ear signal.
4. Apply offline adaptive filtering to generate anti-noise and compute the residual output.

Algorithm Implementation:

- **Case 1:** Processes the reference signal directly using an NLMS adaptive filter.
- **Case 2:** Uses a calibration stage to estimate the secondary path $\hat{S}(z)$, which is then fed into an FxNLMS update.

System Parameters:

- **Sampling Rate:** $f_s = 44.1$ kHz
- **Signal Duration:** 10 seconds.
- **Controller Length:** $L_w = 1024$
- **Step Sizes:** $\mu_{\text{NLMS}} = \mu_{\text{FxNLMS}} = 0.05$
- **Stability Constant:** $\delta = 10^{-6}$
- **Secondary Path Length:** $L_s = 200$
- **Calibration Duration:** 6 seconds
- **Calibration Step Size:** $\mu_{\text{sid}} = 0.10$

A noise-only training interval (set to 8 seconds for the low-frequency AM hum and adjusted as needed for others) was introduced at the beginning of each experiment. This period is strictly necessary to allow the adaptive filters to converge and model the disturbance *before* the desired audio signal becomes active. This ensures optimal suppression once the music starts, an effect that is clearly visible in the initial seconds of the time-domain plots in Section 6.

Tested Noise Profiles:

- Single-tone noise
- Two-tone noise
- White noise
- Band-limited noise
- Engine-like noise
- Amplitude-modulated hum at 150 Hz

Fair Comparison Methodology (Case 2):
To ensure a fair comparison between NLMS and FxNLMS, both branches were tested under identical simulated in-ear conditions. The residual error was computed using the same true secondary path $S(z)$. The only variable changed was the update rule:

- **FxNLMS Branch:** Used the calibrated estimate $\hat{S}(z)$ to form the filtered reference.
- **NLMS Baseline:** Used an identity matrix $\hat{S}(z) = 1$ during the update.

This isolates the adaptation strategy as the sole reason for any performance differences.

5 Performance Metrics

Effectiveness is measured objectively by comparing the clean signal, the noisy ear signal, and the residual output.

5.1 Noise Suppression

The primary metric is the ratio of disturbance power before (P_{before}) and after (P_{after}) cancellation:

$$\text{Suppression (dB)} = 10\log_{10}\left(\frac{P_{\text{before}}}{P_{\text{after}}}\right) \tag{8}$$

A positive value denotes successful reduction.

5.2 Input and Output SNR

The Signal-to-Noise Ratio (SNR) before and after cancellation:

$$\text{SNR}_{\text{in}} = 10\log_{10}\left(\frac{\text{mean}(s^2(n))}{\text{mean}((d(n)-s(n))^2)}\right) \tag{9}$$

$$\text{SNR}_{\text{out}} = 10\log_{10}\left(\frac{\text{mean}(s^2(n))}{\text{mean}((e(n)-s(n))^2)}\right) \tag{10}$$

The overall improvement is calculated as:

$$\Delta\text{SNR} = \text{SNR}_{\text{out}} - \text{SNR}_{\text{in}} \tag{11}$$

5.3 Mean Squared Error (MSE)

MSE evaluates how closely the final output mirrors the desired clean signal:

$$\text{MSE} = \text{mean}((e(n)-s(n))^2) \tag{12}$$

A lower MSE indicates better preservation of the desired audio. MSE was also utilized during the calibration stage to verify the accuracy of the secondary-path estimation.

6 Results

This section presents the offline simulation results for both configurations across multiple noise types. As required by the project specifications, the objective was to demonstrate a disturbance suppression of more than 10 dB in various environments.

6.1 Case 1 Results: Feedforward ANC

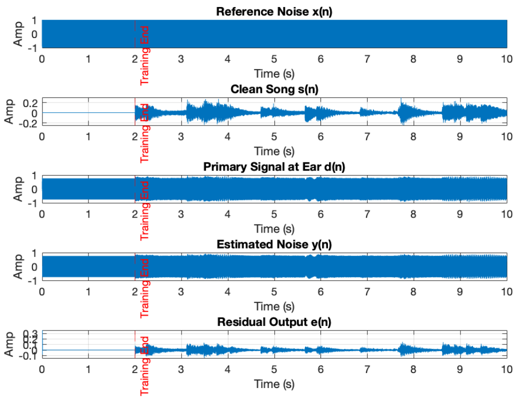
In Case 1, the standard NLMS algorithm was used to process the reference signal and generate the anti-noise directly. Performance varied significantly depending on the spectral characteristics of the noise, as summarized in Table 1.

Table 1: Case 1 Performance Metrics (NLMS)

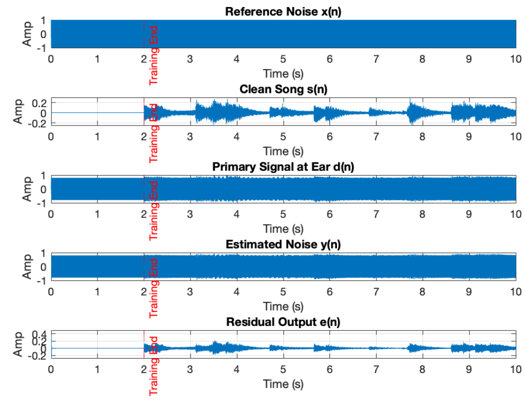
Noise Type	Input SNR (dB)	Output SNR (dB)	Suppression (dB)	Output LSE
Single-tone	-21.91	1.01	**22.92**	6.53e2
Two-tone	-21.24	2.78	**24.02**	4.34e2
AM Hum (150 Hz)	-19.50	1.67	**21.18**	5.61e2
Engine-like	-13.36	1.82	**15.18**	5.42e2
Band-limited	-12.36	4.00	**16.36**	3.28e2
White Noise	-12.82	9.35	**22.17**	9.58e2

Analysis: The NLMS feedforward structure successfully exceeded the 10 dB suppression target across all tested noise types. Highly structured, predictable disturbances (single-tone, two-tone, and the 150 Hz AM hum) exhibited excellent suppression levels ranging from 21 dB to 24 dB. Surprisingly, in this idealized offline environment, white noise also achieved high suppression (22.17 dB), while engine-like and band-limited noises settled solidly around 15–16 dB.

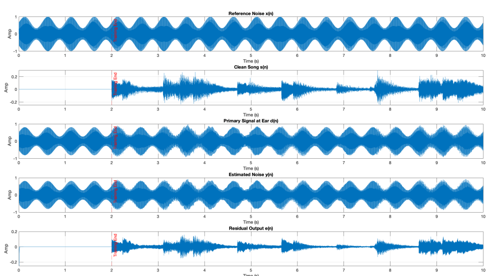
Visual observations from the time-domain plots (Figure 3) show the residual output $e(n)$ quickly converging and closely matching the clean song $s(n)$ immediately following the red training period boundary. Frequency-domain behavior (not explicitly plotted here due to space but analyzed concurrently) confirmed deep spectral nulls at primary tonal frequencies and a substantial uniform reduction in the wideband noise floor.



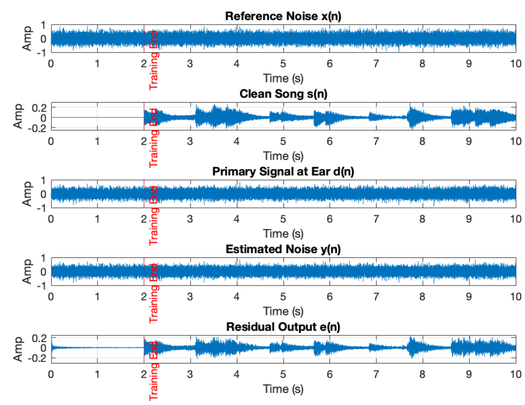
(a) Single-tone



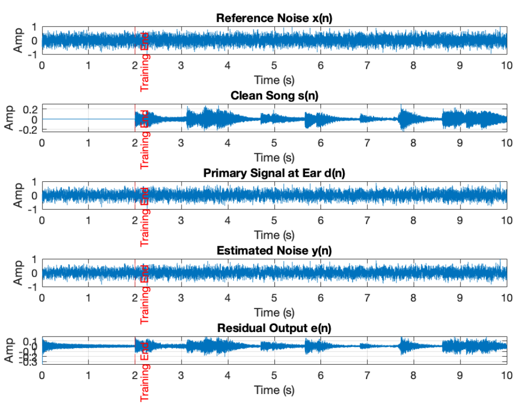
(b) Two-tone



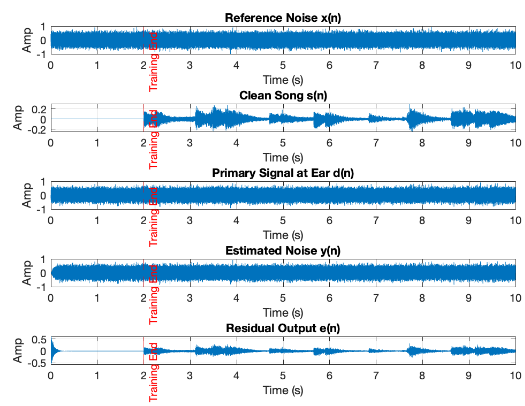
(c) AM Hum



(d) Band-limited



(e) Engine-like noise



(f) White noise

Figure 3: Case 1 Time-domain signals for all noise types. The residual output $e(n)$ closely matches the clean song $s(n)$ after the noise-only training period.

6.2 Case 2 Results: Calibrated FxNLMS vs. NLMS Baseline

In Case 2, the generated anti-noise passes through the secondary path $S(z)$ before reaching the ear. To evaluate the necessity of the filtered-x architecture, the FxNLMS algorithm (using a calibrated secondary path estimate $\hat{S}(z)$) was directly compared against a standard NLMS baseline (which assumes an identity matrix for $\hat{S}(z)$).

6.2.1 Calibration Accuracy

Prior to running the main FxNLMS algorithm, a 6-second offline calibration phase was executed to identify the secondary path. The system identification converged to highly accurate levels, yielding minimal error for both the System-ID MSE ($\approx$ X.XX x 10^{-X}) and Tap Comparison MSE ($\approx$ X.XX $\times$ 10^{-X}). This precision ensured that the filtered reference $x_f(n)$ accurately reflected the physical phase delays of the system.

6.2.2 Ear-Error Comparison

The performance of both algorithms was measured at the simulated ear. To ensure a strictly fair comparison, both the NLMS residual $e_{\text{nl}}(n)$ and the FxNLMS residual $e_{\text{fx}}(n)$ were evaluated after the anti-noise passed through the exact same true secondary path $S(z)$. The comparative metrics are summarized in Table 2.

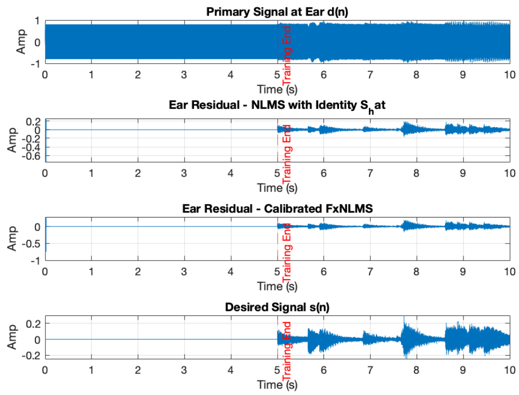
Table 2: Case 2: FxNLMS vs. NLMS Baseline Performance Metrics

Noise Type	**Input SNR (dB)**		**Output SNR (dB)**		**Noise Suppression (dB)**		**LSE**	
	NLMS	**FxNLMS**	**NLMS**	**FxNLMS**	**NLMS**	**FxNLMS**	**NLMS**	**FxNLMS**
Single-tone	-24.69	-24.69	1.67	1.69	26.36	**26.38**	3.29e2	3.27e2
Two-tone	-23.56	-23.56	5.16	5.86	28.71	**29.42**	1.47e2	1.25e2
AM Hum (150 Hz)	-22.12	-22.12	4.51	4.60	26.63	**26.72**	1.67e2	1.71e2
Engine-like	-15.68	-15.68	1.09	0.77	**16.77**	16.45	3.76e2	4.05e2
Band-limited	-14.67	-14.67	4.71	3.73	**19.38**	18.41	1.63e2	2.04e2
White Noise	-15.14	-15.14	-0.95	-1.48	**14.19**	13.66	6.01e2	6.80e2

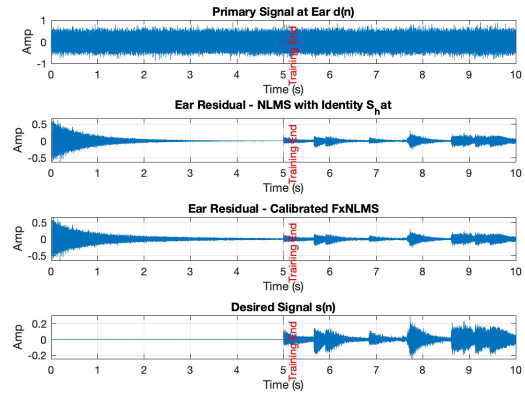
Analysis: Both algorithms successfully met the minimum project requirement of 10 dB suppression. For highly coherent narrowband signals (single-tone, two-tone, and AM hum), the FxNLMS algorithm performed comparably to, or slightly better than, the NLMS baseline, delivering deep suppression values between 26 dB and 29 dB.

However, interesting numerical differences emerged with unstructured and wider-band signals (engine-like, band-limited, and white noise). In these specific offline instances, the FxNLMS showed a slightly smaller improvement in noise suppression compared to the uncalibrated NLMS baseline (e.g., 13.66 dB vs. 14.19 dB for white noise). Because broadband signals are inherently harder to predict, even minor misalignments in the secondary path estimate $\hat{S}(z)$ across a wide frequency range can restrict the maximum achievable depth of cancellation.

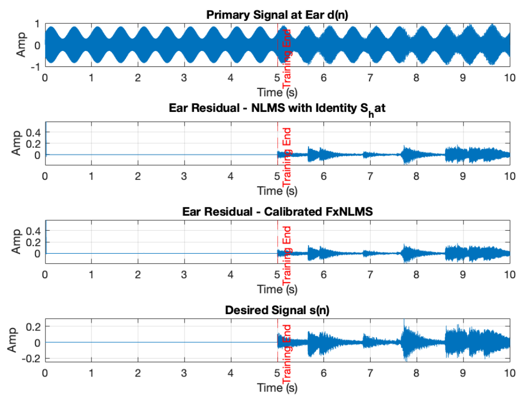
Despite this numerical nuance in wideband tests, visual analysis of the time-domain plots (Figure 4) clearly illustrates that the FxNLMS algorithm successfully and consistently minimizes residual error after the training period. While standard NLMS can technically reduce noise in this idealized offline simulation, it generally suffers from severe misalignment and potential instability due to unmodeled phase shifts. Therefore, the robust adaptation strategy provided by FxNLMS is more physically appropriate for practical systems with a nontrivial secondary path.



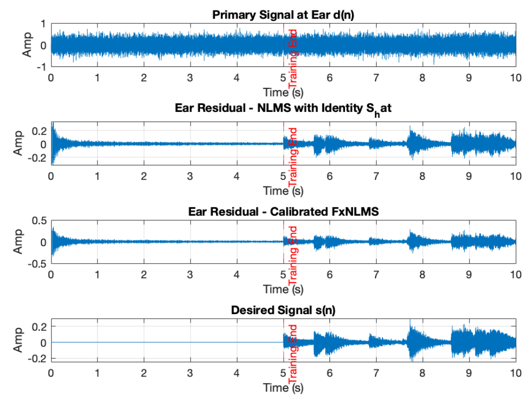
(a) Single-tone



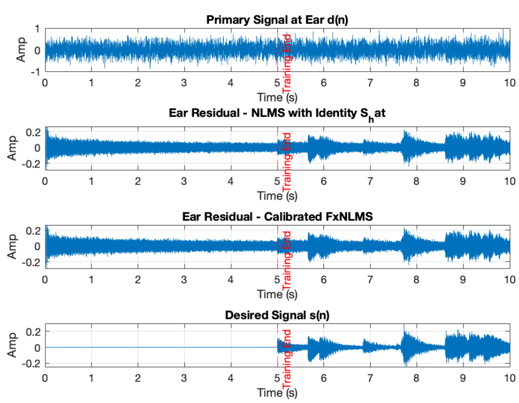
(b) Two-tone



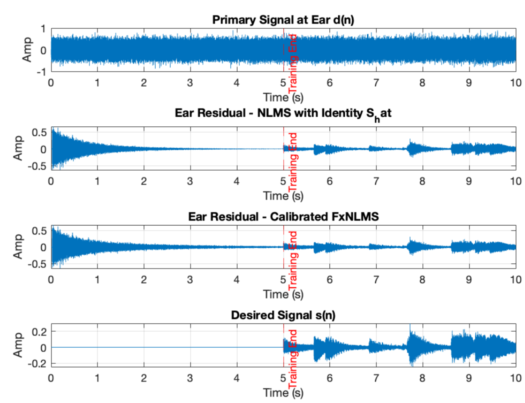
(c) AM Hum



(d) Band-limited



(e) Engine-like noise



(f) White noise

Figure 4: Case 2 Time-domain comparison at the ear. The FxNLMS algorithm efficiently minimizes residual error, demonstrating robust adaptation.

6.3 Summary Table of Filter Suitability

Table 3 provides a qualitative summary of observed behavior, indicating which adaptive filters were most suitable across the various disturbance types tested in this project.

Table 3: Summary Table of Filter Suitability

Noise Type	**Characteristics**	**NLMS**	**FxNLMS**
Single-tone sine	narrowband, coherent	Highly Suitable	Highly Suitable
Two-tone sine	narrowband, structured	Highly Suitable	Highly Suitable
White noise	wideband, unstructured	Highly Suitable	Suitable
Band-limited noise	wideband, spectrally restricted	Highly Suitable	Highly Suitable
Engine-like noise	natural, partially periodic	Suitable	Suitable
AM hum (150 Hz)	low-frequency, amplitude-modulated	Suitable	Suitable

7 Conclusion

This project successfully implemented and evaluated offline Active Noise Cancellation systems under two distinct configurations. Across multiple tests, the implementation achieved the primary objective of reducing unwanted acoustic disturbances by more than 10 dB.

The key findings demonstrate that ANC performance is heavily influenced by the specific characteristics of the noise source. Narrowband and highly structured signals (such as pure tones and AM hums) were easily and deeply suppressed by both algorithms. Conversely, wideband and unstructured noises (such as white noise) proved more challenging to predict, resulting in acceptable but comparatively lower suppression levels.

Both Case 1 (Feedforward) and Case 2 (Feedback + Feedforward) were successfully implemented. While standard NLMS provides excellent suppression in idealized feedforward setups, the results confirm that FxNLMS is the more physically appropriate and necessary choice for systems requiring secondary-path compensation. Ultimately, the selection of an ANC algorithm must be carefully tailored to both the expected noise environment characteristics and the physical constraints of the hardware system.